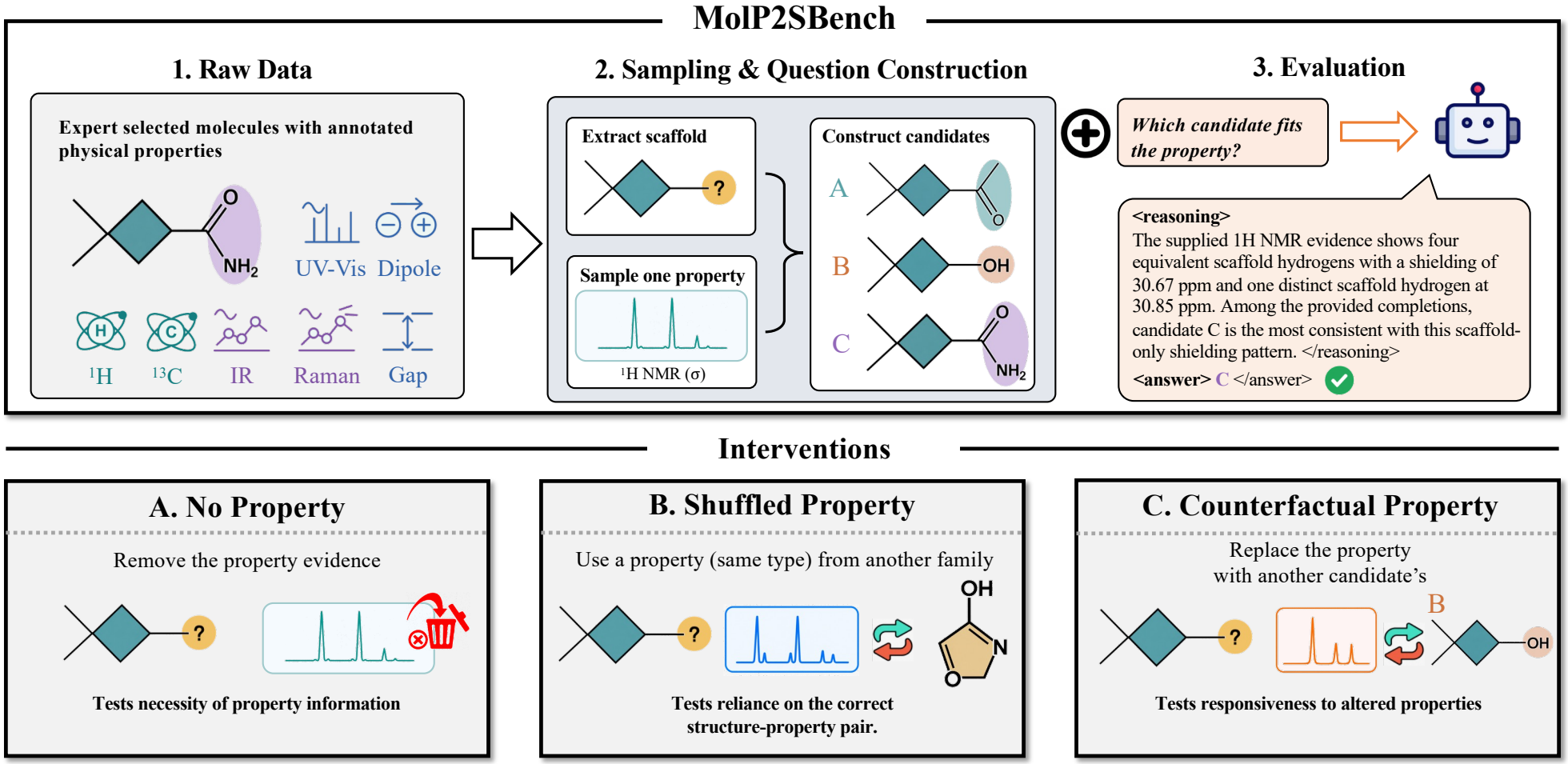

MolP2SBench
1. Raw Data
Expert selected molecules with annotated physical properties
NH2
UV-Vis
Dipole
1H
13C
IR
Raman
Gap
2. Sampling & Question Construction
Extract scaffold
?
Sample one property
1H NMR (σ)
Construct candidates
A
O
B
OH
C
O
NH2
3. Evaluation
Which candidate fits the property?
<reasoning>
The supplied 1H NMR evidence shows four equivalent scaffold hydrogens with a shielding of 30.67 ppm and one distinct scaffold hydrogen at 30.85 ppm. Among the provided completions, candidate C is the most consistent with this scaffold-only shielding pattern. </reasoning>
<answer> C </answer>
Interventions
A. No Property
Remove the property evidence
?
Tests necessity of property information
B. Shuffled Property
Use a property (same type) from another family
?
OH
N
O
Tests reliance on the correct structure-property pair.
C. Counterfactual Property
Replace the property with another candidate's
?
B
OH
Tests responsiveness to altered properties